BOX PRODUCT SURJECTIONS WITHOUT NOETHERIAN HYPOTHESES

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Editorial note. 'We urge the reader to skip the proof.' is a stylistic aside, not a disclaimer; the limit argument is given in full.

Allowed standard prerequisites (editorial).

- Noetherian box surjection. lemma-on-product L26469 (carded this batch). Base case after descent.
- Approximation. limits-proposition-approximate, limits-lemma-descend-modules-finite-presentation, perfect-lemma-resolution-property-descends (cross-chapter, named). Descent of F, the module, and the resolution property.

Source reference. Original lemma 0FZ0: sources/stacks-equiv/pinned-original-source.tex, source lines 1750–1750. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01ZR: focused-seeds/geometry/S9-eqv-on-product-general/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01ZA: focused-seeds/geometry/S9-eqv-on-product-general/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0F8B: focused-seeds/geometry/S9-eqv-on-product-general/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 086M: focused-seeds/geometry/S9-eqv-on-product-general/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:equiv.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let R be a ring. Let X, Y be quasi-compact and quasi-separated schemes over R having the resolution property. For any finite type quasi-coherent $\mathcal{O}_{X\times_R Y}$-module $\mathcal{F}$ there exist a surjection $\mathcal{E}\boxtimes\mathcal{G}\to\mathcal{F}$ where $\mathcal{E}$ is a finite locally free $\mathcal{O}_X$-module and $\mathcal{G}$ is a finite locally free $\mathcal{O}_Y$-module.*

Proof. Follows from Lemma 0FZ0 by a limit argument. We urge the reader to skip the proof. Since $X\times_R Y$ is a closed subscheme of $X\times_{\mathbf{Z}} Y$ it is harmless if we replace R by $\mathbf{Z}$. We can write $\mathcal{F}$ as the quotient of a finitely presented $\mathcal{O}_{X\times_R Y}$-module by Properties, Lemma 086M. Hence we may assume $\mathcal{F}$ is of finite presentation. Next we can write $X = \lim X_i$ with X_i of finite presentation over $\mathbf{Z}$ and similarly $Y = \lim Y_j$, see Limits, Proposition 01ZA. Then $\mathcal{F}$ will descend to $\mathcal{F}_{ij}$ on some $X_i\times_R Y_j$ (Limits, Lemma 01ZR) and so does the property of having the resolution property (Derived Categories of Schemes, Lemma 0F8B). Then we apply Lemma 0FZ0 to $\mathcal{F}_{ij}$ and we pullback. □